

Problem 2

Problem 2

Problem 1 A table of values for $f(x)$ and $f'(x)$ is shown below:

x	$f(x)$	$f'(x)$
1	3	4
2	2	3
3	4	5
4	6	3

- Evaluate the limit $\lim_{x \rightarrow 2} \frac{f(x^2) - 6}{x - 2}$. **EXPLAIN.**

Explanation. If we set $g(x) = f(x^2)$, this limit is $\lim_{x \rightarrow 2} \frac{g(x) - g(2)}{x - 2}$. In other words, we're being asked to evaluate $\frac{d}{dx} f(x^2)$ at $x = 2$. By the Chain Rule, $\frac{d}{dx} f(x^2) = 2x f'(x^2)$. When evaluated at $x = 2$ this is $2(2)f'(4) = 4(3) = 12$. The value of this limit is 12.

Our full explanation is then: $\lim_{x \rightarrow 2} \frac{f(x^2)}{x - 2} = \left[\frac{d}{dx} f(x^2) \right]_{x=2}$ by the limit definition of the derivative. Our calculation above shows that this equals 12.

- Evaluate $\frac{d}{dx} f(f(x))$ at $x = 3$.
 - 6
 - 25
 - 5
 - 15
 - DNE
 - None of the previous answers.

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Explanation. The answer is (d):

$$\begin{aligned}\left[\frac{d}{dx}f(f(x))\right]_{x=3} &= f'(f(3)) \cdot f'(3) \\ &= f'(4) \cdot 5 \\ &= 3 \cdot 5 = 15\end{aligned}$$